**Laboratory work 5**

**We continue to work with the database from the previous laboratory works.**

**Take a full-page screenshot that covers the code and results of each task.**

**Tasks**:

1. Add a CHECK constraint to passenger table to provide that passengers must be at least 10 years old.

   Answer:

```
alter table passengers
ADD constraint age_chack
check (extract(year from AGE(date_of_birth))>10);
```

2. Add a CHECK constraint to accept values in booking price not more than 50000tg and less than 0tg.

   Answer:

```
alter table booking
ADD CONSTRAINT check_price
check (price between 0 and 50000);
```

3. Add a CHECK constraint to accept the luggage weight between 1 and 23 kg.

   Answer:

```
alter table baggage
ADD constraint check_baggage_kg
check (weight_in_kg between 1 and 23);
```

4. Add a CHECK constraint to ensure that all values in airport_name must have at least 10 characters

   Answer:

```
alter table airport
ADD constraint check_airport_name_langht
check (length(airport_name)>=10);
```

5. Add UNIQUE constraint to some columns in each table in database.

   Answer:

```
alter table booking
ADD constraint booking_unq unique (passenger_id);
alter table airport
ADD constraint airport_unq unique (airport_name);
alter table airline
ADD constraint airline_unq unique (airline_code);
alter table passengers
ADD constraint passengers_unq unique (passport_number);
alter table security_check
ADD constraint  security_check_unq unique (passenger_id);
alter table baggage_check
ADD constraint baggage_check_unq unique (booking_id,passenger_id);
alter table baggage
ADD constraint baggage_unq unique (booking_id);
alter table boarding_pass
ADD constraint boarding_pass_unq unique (booking_id);
alter table booking_flight
ADD constraint booking_flight_unq unique (booking_id,flight_id);
alter table flights
ADD constraint flights_unq unique (airline_id);
```

6. Add a CHECK constraint to ensure that male passengers must be at least 18 years old and female passengers must be 19 years old.

   Answer:

```
alter table passengers
ADD constraint check_year_of_gender
check ((gender = 'Male' and extract(year from AGE(date_of_birth))>=18)
or (gender='Female' and extract(year from AGE(date_of_birth))>=19));
```

7. Add a CHECK constraint to add rule as follow (use column country_of_citizenship):
   - Passengers from Kazakhstan must be at least 18 years old.
   - Passengers from France must be at least 17 years old.
   - Passengers from other countries must be at least 19 years old.

   Answer:

```
alter table passengers
ADD constraint national_check
check (
    (country_of_citizenship = 'Kazakhstan' and extract(year from
AGE(date_of_birth))>=18)
    or
    (country_of_citizenship = 'France' and extract(year from
AGE(date_of_birth))>=17)
    or
    (country_of_citizenship not in ('France','Kazakhstan') and extract(year
from AGE(date_of_birth))>=19)
    );
```

8. Add a ticket_discount column to table booking and a CHECK constraint to apply some discount based on ticket price and created time:
    - the constraint applies a 5% discount for tickets created after 2024-01-01,
      and 10% discount for tickets created before 2024-01-01.

    Answer:

```
UPDATE booking
set ticket_discount =
CASE
    when created_at>= DATE '2024-01-01' THEN ROUND(price *0.05)
    when created_at< DATE '2024-01-01' THEN ROUND(price *0.1)
END;
```

```
alter table booking
ADD column ticket_discount NUMERIC(5,2)
check (
    (created_at >= DATE '2024-01-01' and ticket_discount = ROUND(price *
0.05))
    or
    (created_at < DATE '2024-01-01' and ticket_discount = ROUND(price *0.1))
    );
```

9. Modify the booking table so that the booking status column has a default value of 'Confirmed'. Then insert a new booking without specifying booking status.

    Answer:

```
ALTER TABLE booking
ALTER COLUMN status SET DEFAULT 'Confirmed';
```

**10.** Modify the appropriate date column in the database so that when a new record is inserted, the date is automatically set to the current date.

Answer:

```
ALTER table booking
alter column created_at set default current_timestamp
```